Research Highlights

- **The limit is observational equivalence, not model capacity.** Of 13,360 Codeforces submissions, 1,504 (11.3%) report 0 ms, 0 KB and 0 tests passed while spanning three verdicts — a failed compilation, a crash on the first test and a wrong answer on the first test leave identical traces. No metadata-only method can separate them; the argument is model-free and bounds Compilation Error at F1 ≈ 0.77 (our model attains 0.770, with all 261 of its CE predictions inside that cell). A second signature (31 ms, 0 KB, 1 test) holds 13.0% of the corpus and bounds the Wrong Answer $\leftrightarrow$ Runtime Error distinction.

- **Which verdicts a submission log can carry, and at what cost.** Time Limit Exceeded (F1 0.99) and Memory Limit Exceeded (0.95) are fully recoverable because the platform's own thresholds define them; Compilation Error is recoverable only up to its equivalence cell; Runtime Error is not recoverable (F1 0.33, 94 of 131 errors reported as Wrong Answer). Together the execution-encoded classes account for 18.5% of submissions, and 87.8% of the test partition lies in classes classified at F1 ≥ 0.95.

- **A comparison that separates the gap from the ranking.** Gradient Boosting reaches 91.77% (balanced accuracy 0.80), leading DeepSeek-V3 (three-shot, 76.65%) by 15.1 pp, with both effectively free at classroom scale (supervised inference below \$0.000001 per thousand predictions, API about \$0.01), with a five-field variant at 91.69% so the comparison is like-for-like on information. DeepSeek-V3's accuracy is indistinguishable from always answering Wrong Answer (76.61%), yet it recovers every compilation error at 32% precision — it has generalized the platform encoding from a worked example but has no calibrated prior to trade recall for precision inside the equivalence cell. The two tree ensembles are a null result ($\chi^2 = 0.00$, $p = 1.00$) and are reported as such.

- **A leakage finding worth reusing.** A submitter-history aggregate — the standard first feature in grade prediction — appears to add 3.7 pp only because each row's own label leaks into its own predictor; computed without look-ahead the gain shrinks to a nominal 0.67 pp that does not survive correction for multiple comparisons, and it is exactly zero for new submitters. The field is excluded; all six retained fields are computable at first contact with a submission, so the pipeline carries no cold-start penalty. Execution time nonetheless dominates (7.3 pp when removed; 81.5% of Gini importance).

- **Open data, code, and a single reproduction script.** 13,360 submissions with the six public fields; `pipeline_authoritative.py` regenerates every number, table and figure in one run; submitter-disjoint replication preserves both ranking and boundary (GB 93.13%). Available at: `https://github.com/huwenbin123huwenbin/programming-error-classification`